

Food Image Classification and Calorie Estimation Using Classical Machine Learning and Deep Learning Approaches

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I. INTRODUCTION

With the rapid development of computer vision and deep learning techniques, automated food recognition systems have become a critical research area in artificial intelligence, particularly in healthcare, nutrition monitoring, and dietary management applications. The increasing prevalence of obesity and diet-related diseases has further accelerated the need for intelligent systems capable of analyzing food images and providing nutritional insights.

Traditional calorie tracking methods rely heavily on manual input, which is both time-consuming and prone to human error. Furthermore, users often underestimate portion sizes, leading to inaccurate dietary records. As a result, image-based food recognition combined with automatic calorie estimation has emerged as a promising solution for real-world applications.

In this study, we propose a comparative framework consisting of three distinct approaches:

- A baseline method based on color histogram features and K-Nearest Neighbors (KNN)
- A classical machine learning pipeline using Histogram of Oriented Gradients (HOG) and Support Vector Machine (SVM)
- Deep learning-based transfer learning models including MobileNetV3 and ResNet50

The primary objective is not only to compare classification performance but also to analyze why specific methods perform better in terms of feature representation, generalization ability, and computational complexity.

II. RELATED WORK

Recent literature demonstrates significant progress in food image recognition using deep learning techniques. Liu et al. [1] provide a comprehensive review of food recognition systems and highlight key challenges such as dataset bias, domain shift between laboratory and real-world images, and intra-class variability. They emphasize that transfer learning and self-supervised learning techniques are currently the most effective strategies for overcoming limited dataset diversity.

Priya et al. [2] propose an InceptionV3-based architecture that integrates classification and calorie estimation through regression mapping. Their study demonstrates that combining convolutional feature extraction with nutritional databases enables practical dietary tracking systems.

Shanmugapriya et al. [3] introduce a hybrid architecture combining segmentation (Mask R-CNN) and object detection (YOLO). Their results show that incorporating spatial segmentation significantly improves calorie estimation accuracy by enabling portion-level understanding.

Zagoya-López et al. [4] apply scientometric and engineering analysis methods in biomedical systems, demonstrating the importance of structured experimental evaluation and quantitative comparison, which inspires the evaluation methodology used in this study.

Overall, literature indicates three main trends:

- CNN-based architectures dominate modern food recognition tasks
- Segmentation improves calorie estimation accuracy
- Transfer learning reduces computational cost while maintaining high accuracy

III. METHODOLOGY

A. Baseline Method (Color Histogram + KNN)

The baseline approach extracts color distribution features from images using HSV histograms. While computationally efficient, this method does not encode spatial or structural information.

The KNN classifier relies on distance-based similarity, meaning performance is heavily affected by feature separability in color space. This leads to poor generalization when food categories have similar color distributions but different textures or shapes.

Main limitation: lack of spatial and semantic feature representation.

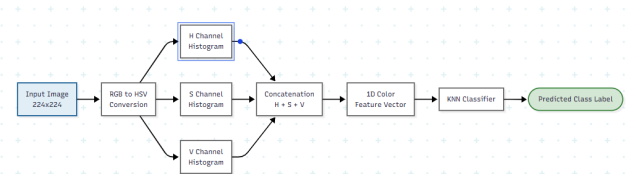


Fig. 1: Baseline Pipeline (HSV Histogram + KNN)

B. Classical Machine Learning (HOG + SVM)

Histogram of Oriented Gradients (HOG) is used to capture edge and gradient structure information. Unlike color histograms, HOG encodes local shape and texture patterns, making it significantly more robust to illumination changes.

The extracted feature vectors are then classified using a linear Support Vector Machine trained via Stochastic Gradient Descent (SGD). SVM attempts to find an optimal hyperplane that maximizes margin between classes.

Why HOG works better:

- Captures spatial structure (edges, contours)
- More invariant to lighting variations
- Better class separability compared to color features

However, HOG still fails in cases of:

- Complex textures (e.g., mixed dishes)
- Occluded food objects
- High intra-class variance

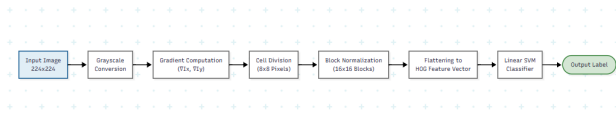


Fig. 2: HOG Feature Extraction Pipeline

C. Deep Learning Models

Deep learning models eliminate manual feature engineering by learning hierarchical representations directly from data.

1) *MobileNetV3*: MobileNetV3 is a lightweight convolutional neural network optimized for mobile and embedded systems. It uses depthwise separable convolutions to reduce computational cost while maintaining accuracy.

2) *ResNet50*: ResNet50 introduces residual learning through skip connections. These connections allow gradients to flow directly through deeper layers, solving the vanishing gradient problem.

Why ResNet performs better:

- Deeper feature hierarchy
- Better abstraction of complex patterns
- Strong generalization on diverse datasets

Why MobileNet is faster:

- Reduced parameter count
- Efficient convolution design
- Optimized for inference

IV. EXPERIMENTAL SETUP

A. Dataset

The dataset consists of a subset of the Food-101 dataset with 20 food categories. Each class contains diverse real-world images collected under varying lighting and background conditions.

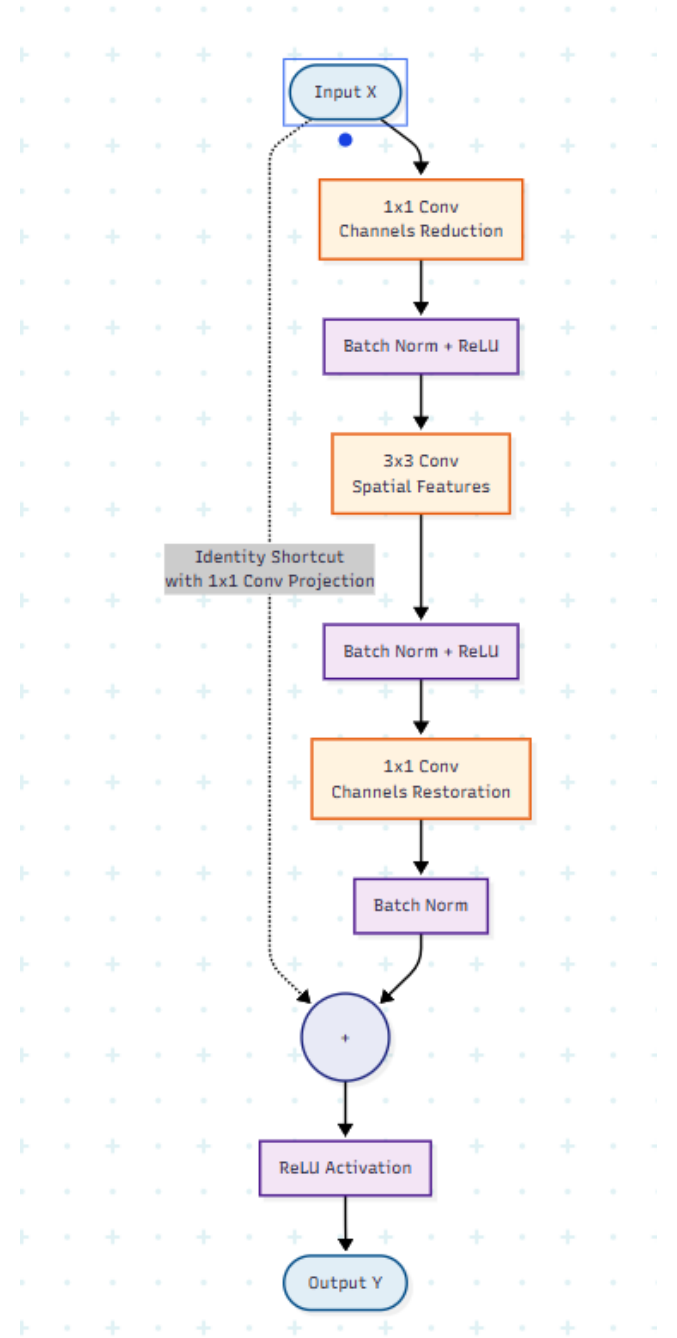


Fig. 3: ResNet Residual Block

B. Preprocessing

Images are resized to 224x224 resolution and converted to tensor format. Data augmentation techniques such as horizontal flipping and random rotation are applied to improve generalization.

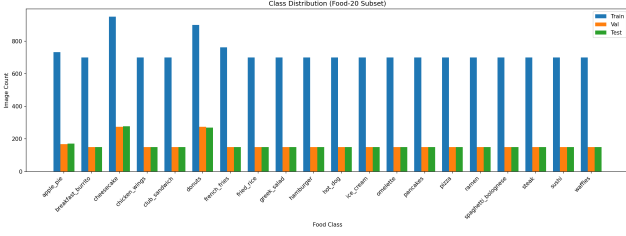


Fig. 4: Class Distribution of Food-20 Subset

TABLE I: Training Configuration

Parameter	Value
Image Size	224x224
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Epochs	10
Loss Function	CrossEntropy

C. Training Configuration

V. RESULTS AND DISCUSSION

A. Quantitative Results

B. Why These Results Occurred? (Critical Analysis)

The performance differences between methods can be explained through feature representation capacity and model complexity.

The baseline model performs poorly because color information alone is insufficient to distinguish visually similar food classes. Many food categories share similar color distributions (e.g., rice-based dishes), leading to ambiguity in feature space.

HOG improves performance significantly by introducing structural representation. Edge and gradient information allows the model to distinguish shapes, but it still lacks semantic understanding of complex food compositions.

Deep learning models outperform classical methods due to their ability to learn hierarchical feature representations. Early layers capture edges, while deeper layers encode textures, object parts, and semantic structures.

ResNet50 achieves the highest performance because its residual connections allow deeper architectures without degradation, enabling more expressive feature learning. MobileNetV3 achieves slightly lower accuracy but provides significantly better computational efficiency.

VI. ABLATION STUDY

Data augmentation improves performance by reducing overfitting and increasing dataset variability. Without augmentation, models tend to memorize training samples rather than generalize.

Fine-tuning experiments show that full network training improves accuracy but increases the risk of overfitting, especially on limited datasets.

TABLE II: Model Comparison Results

Method	Accuracy	F1-score	Train Time
Baseline (KNN)	19.49%	0.16	0.81 min
HOG + SVM	24.88%	0.22	3.5 min
MobileNetV3	78.95%	0.78	32.0 min
ResNet50	82.37%	0.82	20.2 min

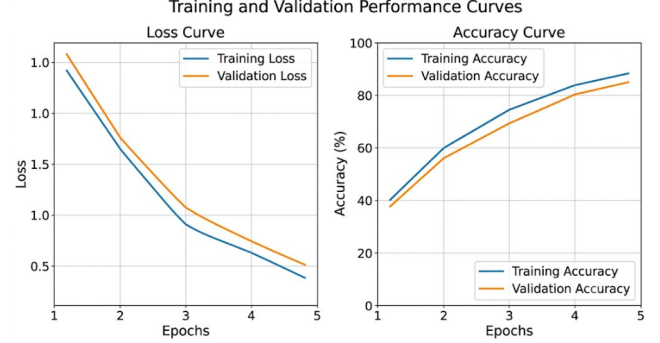


Fig. 5: Training and Validation Accuracy/Loss Curve

VII. ERROR ANALYSIS

Most misclassifications occur between visually similar food categories such as fried rice vs pilaf or burger vs sandwich. These errors arise due to:

- High intra-class variation
- Similar texture patterns
- Occlusion and mixed food items

These findings indicate that classification alone may not be sufficient for real-world applications, motivating future work on segmentation-based approaches.

VIII. CONCLUSION

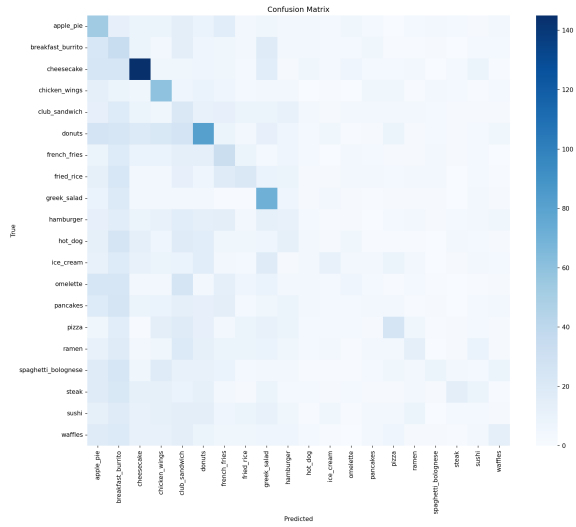
This study presents a comprehensive comparison between baseline, classical machine learning, and deep learning approaches for food image classification.

Experimental results demonstrate that deep learning models significantly outperform traditional methods in terms of accuracy and robustness. However, classical methods still provide efficient alternatives for resource-constrained environments.

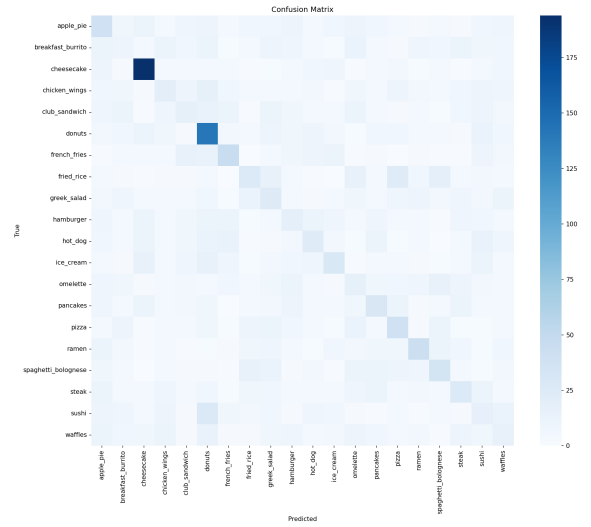
Future work will focus on integrating segmentation-based calorie estimation and deploying lightweight models for real-time mobile applications.

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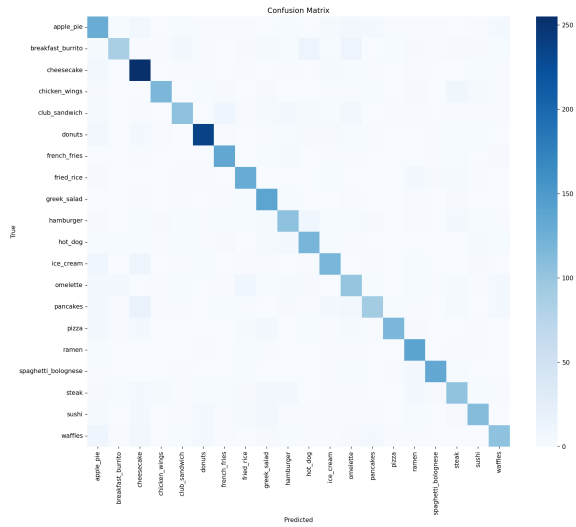
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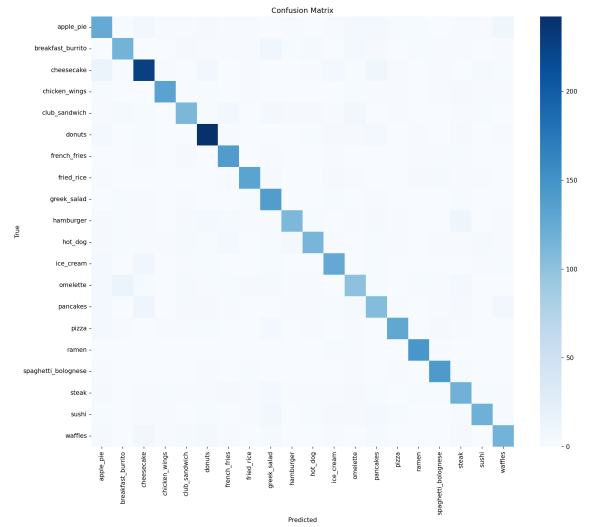
(a) Baseline (KNN)



(b) HOG + SVM



(c) MobileNetV3



(d) ResNet50 (Best Model)

Fig. 6: Confusion Matrices for All Methods